

MBA Fastrack Batch 2025

QUANTITATIVE APTITUDE

Logarithm

DPP: 4

Q1 Solve the following logarithmic equation for x:

$$\log_3(x^2 - 4x + 4) - \log_3(x - 2) = 1$$

- (A) 1 (B) 2
(C) 5 (D) Both B and C

Q2 Find the real root of the equation $3^{4x} + 243(3^{2x})^{-5} - 20 = 0$.

- (A) $\frac{\log_3 7}{2}$
(B) $\log_3 5$
(C) $\frac{\log_3 4}{2}$
(D) $\log_3 25$

Q3 If $\log_7(a) + \log_7(b) = \log_7(5)$ and $\log_3(a) - \log_3(b) = \log_3(2)$, find the value of $(a + b)^2 + (a - b)^2$.

- (A) 25 (B) 50
(C) 75 (D) 100

Q4 If $\log_2(\log_{64} x + \frac{1}{3} + 8^x) = 3x$, then find the value of x^{-1} .

- (A) 2 (B) 3
(C) 4 (D) 5

Q5 Let a, b, and c be positive integers such that $\log_{10}(a) = 2$, $\log_{10}(bc) = 4$, and $\log_{10}(ab) = 3$. What is the value of c?

- (A) 10 (B) 100
(C) 1000 (D) 10000

Q6 If $\log_{256}(16\log_2(1 + \log_6(3 + 3\log_3 x))) = \frac{1}{2}$, then find the value of x.

- (A) 1 (B) 3
(C) 5 (D) 7

Q7 If the solution set of the inequation $(\log_3 x)^4 - 35(\log_3 x)^2 + (\log_3 x + 15)^2 < 30(\log_3 x)$ is $(m, n) \cup (a, b)$, where a, b are positive integers, then find the value of $\frac{1}{2}(b - a)$.

- (A) 105 (B) 106
(C) 107 (D) 108

Q8 If $(\log_{10} x + 4)^3 + (\log_{10} x - 1)^3$, such that $= (2\log_{10} x + 3)^3$, 'x' is an integer, then the number of values that 'x' can assume is:

Q9 If $5^{\log_a \{\log_b(\log_c x)\}} = 1$ and $\log_b \{\log_c (\log_a x)\} = 0$, then find the value of $(\frac{1}{b} + \frac{1}{c})$.

- (A) $\log_a ac$ (B) $\log_{ac} a$
(C) $\log_{ac} ac$ (D) $\log_{ca} ac$

Q10 If $x^1 + \log_{2012} x = 2012x$, then how many real values of 'x' can be obtained?

- (A) 1 (B) 2
(C) 3 (D) 4

Q11 If $\log_{81}(\log_5 x) + \log_{\frac{1}{81}}(\log_5 \frac{1}{y}) = 0$ and $4x^2 = 41 - 4y^2$, then find the value of $2(x + y)$.

- (A) 0 (B) 7
(C) -7 (D) Either 7 or -7

Q12 If $\log_{81}(\log_5 x) + \log_{\frac{1}{81}}(\log_5 \frac{1}{y}) = 0$, then find the value of xy.

- (A) 1
(B) 2
(C) 3



(D) $\frac{1}{2}$

Q13 If $(x-2)^{\log_2 x^2 - 3\log_x 8} = \frac{1}{(x-2)^{17}}$ then find the number of real roots of the equation.

Q14 Let $\log_2(x^2 - 5x + 7) - \log_2(x^2 + 2x + 13) > 2 - \log_2 12$, then the smallest positive integer value that 'x' can take is:

- (A) 7 (B) 8
(C) 9 (D) 10

Q15 The equation $\log_a b^a = \log_b a^{4b} = 20$ satisfies all real numbers a and b. Find ab.

Q16 $\frac{1}{\log_2 x} + \frac{1}{\log_4 x} + \frac{1}{\log_8 x} + \dots + \frac{1}{\log_{8192} x} = ?$
(A) $91 \log_x 2$ (B) $101 \log_x 2$
(C) $81 \log_x 2$ (D) None of these

Q17 How many values of x are possible, if $\log_{2x-3}(7x^3 - 38x^2 + 67x - 37) = 3$.

Q18 If $6^x = 12^y = 72^z$ where x, y and z are non zero real numbers, then find the value of $\frac{x}{z} - \frac{x}{y}$.

Q19 How many integer values of y satisfy the inequality $\log_{0.5}(3y - 4) > \log_{0.25}(64)$?
(A) 2
(B) 3
(C) Infinite
(D) None of the above

Q20 If $x = \sum_{n=1}^{25} \log_{243} 9^n$, find 5x.

Q21 If $\log_{(x-2)}(x^3 - 8x^2 + 22x - 20) = 2$, how many integral values of x are possible?
(A) 0 (B) 1
(C) 2 (D) 3

Q22 If $\log_5 m = \left(\frac{1}{\log_3 5} + \frac{1}{\log_7 5} + \frac{1}{\log_{19} 5}\right)$, and $\log_{10} \log_2(5^{\sqrt{5x+1}} + m) = 1$, then find the value of x.

Q23 If $\log_7(x-1) - 2\log_{7^{-2}}(x+2) + \log_7(x-5) < \log_7[31 - x(x-18)]$ then the number of possible integer values of x is:

Q24 The number of solutions to the equation $\log_9(x-2) = \log_3(x-4)$ is:
(A) 0 (B) 1
(C) 2 (D) 3

Q25 If a, b, c are distinct positive real numbers, such that $(\log_b a \cdot \log_c a - 1) + (\log_a b \cdot \log_c b - 1) + (\log_a c \cdot \log_b c - 1) = 0$ what is the value of abc?

Q26 For how many integral values of y is the given expression non negative?
 $\frac{\log_4 y - 3}{4 - \log_3 y}$

Q27 How many real values of x satisfy the following equation
 $\log_x \frac{x^2+9}{x^2-7} = 2$
(A) 0 (B) 1
(C) 2 (D) 3

Q28 The sum of all integer values of x that satisfy $x^2 + 2x - 8 \leq 0$ such that

$$\log_{10} \left[\frac{9x-2 \left(3^{\log_3 x + \log_3 x^{\frac{1}{2}} + \log_3 x^{\frac{1}{4}} + \dots \right)}{7} \right] \text{ is defined}$$

Q29 Find the value of $\frac{3xy}{x+y}$ if $5^x = 11^y = 3025$.

Q30 The number of possible negative integer values of x in $\log_{(x+7)} \left[\frac{x+5}{x+3} \right]$ such that the expression is defined:
(A) 5 (B) 4
(C) 3 (D) 2



Answer Key

Q1 C
Q2 C
Q3 A
Q4 C
Q5 C
Q6 B
Q7 D
Q8 1
Q9 C
Q10 B
Q11 B
Q12 A
Q13 3
Q14 C
Q15 100

Q16 A
Q17 0
Q18 1
Q19 A
Q20 650
Q21 B
Q22 3
Q23 1
Q24 B
Q25 1
Q26 17
Q27 B
Q28 3
Q29 6
Q30 D



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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

To solve the logarithmic equation, we can first apply the properties of logarithms to simplify the equation.

By using quotient rule of logarithms, which states that $\log_a(m) - \log_a(n) = \log_a\left(\frac{m}{n}\right)$, we can rewrite the given equation:

$$\log_3\left(\frac{x^2 - 4x + 4}{x-2}\right) = 1$$

Now, we can use the property of logarithms, which states that if $\log_a(m) = n$, then $a^n = m$:

$$3^1 = \frac{x^2 - 4x + 4}{x-2}$$

Simplifying the equation, we get:

$$3 = \frac{x^2 - 4x + 4}{x-2}$$

Now, let's get rid of the fraction by multiplying both sides of the equation by $(x - 2)$:

$$3(x - 2) = x^2 - 4x + 4$$

Expanding and simplifying the equation, we get:

$$3x - 6 = x^2 - 4x + 4$$

Rearranging the terms to form a quadratic equation:

$$x^2 - 7x + 10 = 0$$

Now, let's try to factor the quadratic equation:

$$(x - 2)(x - 5) = 0$$

This gives us two potential solutions:

$$x - 2 = 0 \Rightarrow x = 2$$

$$x - 5 = 0 \Rightarrow x = 5$$

However, we need to check if these solutions are valid in the original logarithmic equation. It's important to remember that the argument of a logarithm must always be greater than zero.

For $x = 2$, the first term of the original equation becomes $\log_3(0)$, which is undefined.

Therefore, $x = 2$ is not a valid solution.

For $x = 5$, both terms in the original equation are defined:

$$\log_3\left(\frac{5^2 - 4 \times 5 + 4}{5-2}\right) = 1$$

So, $x = 5$ satisfies the original equation.

Video Solution:



Q2 Text Solution:

Given that,

$$3^{4x} + 243(3^{2x-5}) - 20 = 0$$

$$(3^{2x})^2 + 3^5 \times 3^{2x-5} - 20 = 0$$

$$(3^{2x})^2 + 3^{2x} - 20 = 0$$

$$(3^{2x})^2 + 5(3^{2x}) - 4(3^{2x}) - 20 = 0$$

$$3^{2x}(3^{2x} + 5) - 4(3^{2x} + 5) = 0$$

$$(3^{2x} - 4)(3^{2x} + 5) = 0$$

$$3^{2x} = 4, \text{ but } 3^{2x} \neq -5$$

$$\log_3(3^{2x}) = \log_3 4$$

$$2x \log_3 3 = \log_3 4$$

$$2x = \log_3 4$$

$$x = \frac{\log_3 4}{2}$$

Video Solution:



Q3 Text Solution:

We have two equations:

$$\log_7(a) + \log_7(b) = \log_7(5)$$

$$\log_3(a) - \log_3(b) = \log_3(2)$$

Using the properties of logarithms, we can rewrite the equations as follows:

$$\log_7(ab) = \log_7(5)$$



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$$\log_3 \left(\frac{a}{b} \right) = \log_3(2)$$

Since the logarithms are equal, we can equate the arguments:

$$ab = 5 \dots (1)$$

$$\frac{a}{b} = 2 \dots (2)$$

From equation 2, we can find a:

$$a = 2b \dots (3)$$

Substitute the value of a in equation 1:

$$(2b)b = 5$$

$$2b^2 = 5$$

$$b^2 = \frac{5}{2}$$

Now, find the value of a^2 :

$$a^2 = (2b)^2 = 4b^2 = 4 \left(\frac{5}{2} \right) = 10$$

$$(a+b)^2 + (a-b)^2 = 2(a^2 + b^2).$$

$$= 2 \left(10 + \frac{5}{2} \right) = 2 \times \frac{25}{2} = 25$$

Video Solution:



Q4 Text Solution:

Given that,

$$\log_2 \left(\log_{64} x + \frac{1}{3} + 8^x \right) = 3x$$

- $\log_{64} x + \frac{1}{3} + 8^x = 2^{3x}$
- $\log_{64} x + \frac{1}{3} + 8^x = (2^3)^x$
- $\log_{64} x + \frac{1}{3} + 8^x = 8^x$
- $\log_{64} x + \frac{1}{3} = 0$
- $\log_{64} x = -\frac{1}{3}$
- $x = 64^{-\frac{1}{3}}$
- $x = (4^3)^{-\frac{1}{3}} = 4^{-1}$

$$\bullet \quad x^{-1} = 4$$

Video Solution:



Q5 Text Solution:

Given:

$$\log_{10}(a) = 2 \dots (1)$$

$$\log_{10}(bc) = 4 \dots (2)$$

$$\log_{10}(ab) = 3 \dots (3)$$

From the first equation, we can write:

$$a = 10^2 = 100$$

From the third equation, we can write:

$$ab = 10^3$$

Since we know the value of a, we can find the value of b:

$$100 \times b = 10^3$$

$$b = 10$$

Now we can use the second equation:

$$\log_{10}(bc) = 4$$

Substitute the values of b and c:

$$\log_{10}(10 \times c) = 4$$

Using the properties of logarithms:

$$\log_{10}(10) + \log_{10}(c) = 4$$

Since $\log_{10}(10) = 1$:

$$1 + \log_{10}(c) = 4$$

$$\log_{10}(c) = 3$$

$$c = 10^3 = 1000$$

Video Solution:



Q6 Text Solution:



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$$\log_{256}(16\log_2(1 + \log_6(3 + 3\log_3 x))) = \frac{1}{2}$$

$$\Rightarrow 256^{\frac{1}{2}} = 16\log_2(1 + \log_6(3 + 3\log_3 x))$$

$$\Rightarrow 16\log_2(1 + \log_6(3 + 3\log_3 x)) = 16$$

$$\Rightarrow \log_2(1 + \log_6(3 + 3\log_3 x)) = 1$$

$$\Rightarrow (1 + \log_6(3 + 3\log_3 x)) = 2$$

$$\Rightarrow \log_6(3 + 3\log_3 x) = 1$$

$$\Rightarrow 3 + 3\log_3 x = 6,$$

$$\Rightarrow 3\log_3 x = 3$$

Hence $x = 3$.

Video Solution:



Q7 Text Solution:

Given that,

$$(\log_3 x)^4 - 35(\log_3 x)^2 + (\log_3 x + 15)^2 < 30(\log_3 x)$$

- $(\log_3 x)^4 - 35(\log_3 x)^2 + (\log_3 x)^2 + 30(\log_3 x) + (15)^2 < 30(\log_3 x)$
- $(\log_3 x)^4 - 34(\log_3 x)^2 + 225 < 0$
- $(\log_3 x)^4 - 9(\log_3 x)^2 - 25(\log_3 x)^2 + 225 < 0$
- $[(\log_3 x)^2 - 9][(\log_3 x)^2 - 25] < 0$
- $9 < (\log_3 x)^2 < 25$
- $3 < \log_3 x < 5$ or, $-5 < \log_3 x < -3$
- $3^3 < x < 3^5$ or, $3^{-5} < x < 3^{-3}$

- $27 < x < 243$ or,
 $\frac{1}{243} < x < \frac{1}{27}$

Now, since, 'a' and 'b' are positive integers.
Therefore, $a = 27$ and $b = 243$.

$$\text{Then, } \frac{1}{2}(b - a) = \frac{1}{2}(243 - 27) = 108.$$

Video Solution:



Q8 Text Solution:

$$\text{Given that, } (\log_{10} x + 4)^3 + (\log_{10} x - 1)^3 = (2\log_{10} x + 3)^3$$

.... (i)

$$\text{Let } \log_{10} x + 4 = a, \log_{10} x - 1 = b$$

$$\text{Then, } a + b = 2\log_{10} x + 3$$

Therefore, (i), we have

$$a^3 + b^3 = (a + b)^3$$

- $3ab(a + b) = 0$
- $a = 0$, or, $b = 0$ or, $a + b = 0$
- $\log_{10} x + 4 = 0$ or,
 $\log_{10} x - 1 = 0$ or, $2\log_{10} x + 3 = 0$
- $\log_{10} x = -4$ or, $\log_{10} x = 1$ or,
 $\log_{10} x = -\frac{3}{2}$
- $x = 10^{-4}$ or, $x = 10^1$ or, $x = 10^{-\frac{3}{2}}$
- $x = \frac{1}{10000}$ or, $x = 10$ or,
 $x = \frac{1}{10\sqrt{10}}$

As, 'x' is an integer, only 1 value of 'x' is possible which is 10

Video Solution:





Q9 Text Solution:

$$\text{Given that, } 5^{\log_a \{\log_b (\log_c x)\}} = 1$$

$$\Rightarrow 5^{\log_a \{\log_b (\log_c x)\}} = 5^0$$

$$\Rightarrow \log_a \{\log_b (\log_c x)\} = 0$$

$$\Rightarrow \log_b (\log_c x) = a^0 = 1$$

$$\Rightarrow \log_c x = b^1 = b$$

$$\Rightarrow \frac{1}{b} = \frac{1}{\log_c x} = \log_x c \quad \dots\dots (i)$$

$$\text{Also, } \log_b \{\log_c (\log_a x)\} = 0$$

$$\log_c (\log_a x) = b^0 = 1$$

$$\Rightarrow \log_a x = c^1 = c \quad \dots (ii)$$

$$\Rightarrow \frac{1}{c} = \frac{1}{\log_a x} = \log_x a$$

Again, from (ii), we have

$$x = a^c \dots (iv)$$

$$\text{Now, } \left(\frac{1}{b} + \frac{1}{c}\right) = \log_x c + \log_x a$$

$$= \log_x ac$$

$$= \log_{ac} ac \quad [\text{using (iv)}]$$

Hence, option (C) is correct.

Video Solution:



Q10 Text Solution:

Given that

$$x^{1 + \log_{2012} x} = 2012x$$

- $x^1 \cdot x^{\log_{2012} x} = 2012x$
- $x (x^{\log_{2012} x} - 2012) = 0$
- $x \neq 0,$
 $(x^{\log_{2012} x} - 2012) = 0$

- $x^{\log_{2012} x} = 2012$
- $\log_{2012} x = \log_x 2012$
- $\log_{2012} x = \frac{1}{\log_{2012} x}$
- $(\log_{2012} x)^2 = 1$
- $\log_{2012} x = \pm 1$
- $x = 2012^{\pm 1}$
- $x = 2012, \frac{1}{2012}$

Hence, 2 real values of x can be obtained.

Video Solution:



Q11 Text Solution:

Given that,

$$\log_{81} (\log_5 x) + \log_{\frac{1}{81}} \left(\log_5 \frac{1}{y} \right) = 0$$

- $\log_{3^4} (\log_5 x) + \log_{3^{-4}} \left(\log_5 \frac{1}{y} \right) = 0$
- $\frac{1}{4} \log_3 (\log_5 x) - \frac{1}{4} \log_3 \left(\log_5 \frac{1}{y} \right) = 0$



$$\frac{1}{4} \left[\log_3 \frac{\log_5 x}{\log_5 \frac{1}{y}} \right] = 0$$

$$\log_3 \frac{\log_5 x}{\log_5 \frac{1}{y}} = 0$$

$$\frac{\log_5 x}{\log_5 \frac{1}{y}} = 1$$

$$\log_5 x = \log_5 \frac{1}{y}$$

$$x = \frac{1}{y}$$

$$xy = 1$$

Now,

$$4x^2 = 41 - 4y^2$$

- $4x^2 + 4y^2 = 41$
- $x^2 + y^2 = \frac{41}{4}$
- $(x + y)^2 - 2xy = \frac{41}{4}$
- $(x + y)^2 - 2 = \frac{41}{4}$ [Using (i)]
- $(x + y)^2 = \frac{41}{4} + 2$
- $(x + y)^2 = \frac{49}{4}$

Now both 'x' and 'y' must be positive as the argument of a logarithm is always positive

$$\text{Thus, } x + y = \frac{7}{2}$$

$$2(x + y) = 7$$

Video Solution:



Q12 Text Solution:

Given that,

$$\log_{81}(\log_5 x) + \log_{\frac{1}{81}}\left(\log_5 \frac{1}{y}\right) = 0$$

- $\log_{3^4}(\log_5 x) + \log_{3^{-4}}\left(\log_5 \frac{1}{y}\right) = 0$
- $\frac{1}{4} \log_3(\log_5 x) - \frac{1}{4} \log_3(-\log_5 y) = 0$
[Since, $\log_{A^c} B = \frac{1}{c} \log_A B$]
- $\frac{1}{4} \left[\log_3 \left(-\frac{\log_5 x}{\log_5 y} \right) \right] = 0$
- $\log_3 \left(-\frac{\log_5 x}{\log_5 y} \right) = 0$
- $-\frac{\log_5 x}{\log_5 y} = 3^0 = 1$
- $\log_5 x = -\log_5 y$
- $x = \frac{1}{y}$
- $x = \frac{1}{y}$
- $xy = 1$

Video Solution:



Q13 Text Solution:

Given that,

$$(x - 2)^{\log_2 x^2 - 3 \log_x 8} = \frac{1}{(x-2)^{17}}$$

- $(x - 2)^{\log_2 x^2 - 3 \log_x 8} = (x - 2)^{-17}$

Taking log on both sides, we have



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- $(\log_2 x^2 - 3\log_x 8) \log(x-2) = -17 \log(x-2)$
- $\log(x-2) [\log_2 x^2 - 3\log_x 2^3 + 17] = 0$
- $\log(x-2) \left[2\log_2 x - \frac{9}{\log_2 x} + 17 \right] = 0$
- $\log(x-2) \left[2(\log_2 x)^2 + 17\log_2 x - 9 \right] = 0$
- $\log(x-2) \left[2(\log_2 x)^2 + 18\log_2 x - \log_2 x - 9 \right] = 0$
- $\log(x-2) (2\log_2 x - 1)(\log_2 x + 9) = 0$
- $\log(x-2) = 0$ or, $(2\log_2 x - 1) = 0$ or, $(\log_2 x + 9) = 0$
- $x - 2 = 1$ or, $\log_2 x = \frac{1}{2}$ or, $\log_2 x = -9$
- $x = 3$ or, $x = 2^{\frac{1}{2}} = \sqrt{2}$ or, $x = 2^{-9} = \frac{1}{512}$

Hence, the number of real roots of the given equation is 3.

Video Solution:



Q14 Text Solution:

$$\log_2(x^2 - 5x + 7) - \log_2(x^2 + 2x + 13) > 2 - \log_2 12$$

$$\log_2 \frac{x^2 - 5x + 7}{x^2 + 2x + 13} > 2 - \log_2 12$$

Using the property of logarithm, we can rewrite the inequality:

$$\log_2 \frac{x^2 - 5x + 7}{x^2 + 2x + 13} + \log_2 12 - \log_2 4 > 0$$

$$\log_2 \frac{x^2 - 5x + 7}{x^2 + 2x + 13} + \log_2 \frac{12}{4} > 0$$

$$\log_2 \frac{x^2 - 5x + 7}{x^2 + 2x + 13} + \log_2 3 > 0$$

$$\log_2 \frac{3(x^2 - 5x + 7)}{x^2 + 2x + 13} > 0$$

Now, we need to find the range of x values for which the expression inside the logarithm is greater than 1:

$$\frac{3(x^2 - 5x + 7)}{x^2 + 2x + 13} > 1$$

Let's rewrite the inequality:

$$3(x^2 - 5x + 7) > x^2 + 2x + 13$$

(Cross multiplication is allowed since $x^2 + 2x + 13 > 0$ for all real values of x as $D < 0$ here)

Now, simplify:

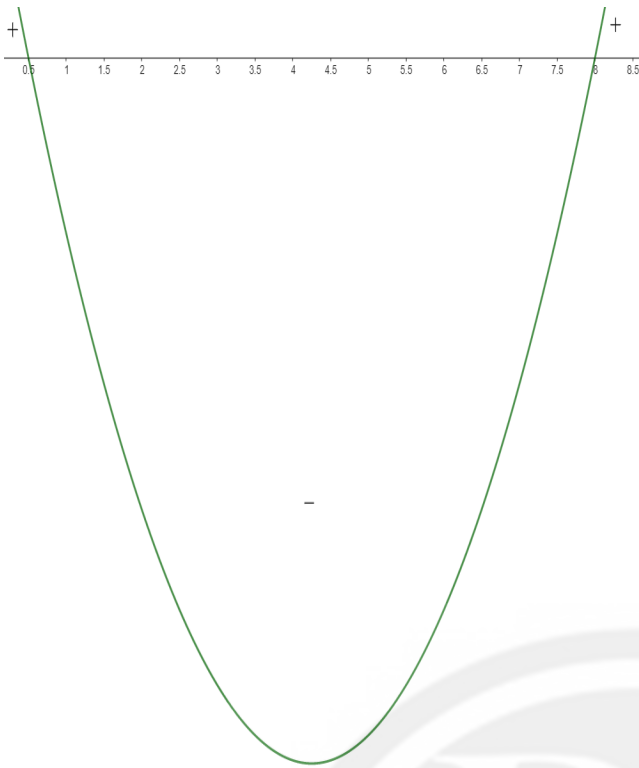
$$2x^2 - 17x + 8 > 0$$

Next, factor the quadratic:

$$(2x - 1)(x - 8) > 0$$

Using the Wavy Curve method, we have





The inequality is satisfied when $x < \frac{1}{2}$ or $x > 8$.

Therefore, the minimum positive integer value of x that can be assumed is 9.

Video Solution:



Q15 Text Solution:

$$a \log_a b = 20 \text{eq (1)}$$

$$4b \log_b a = 20 \text{eq (2)}$$

Multiplying both equation,

$$4ab \log_a b \cdot \log_b a = 400$$

Also, we know that $\log_a b \cdot \log_b a = 1$ (property of Log function)

$$\text{So, } 4ab = 400$$

$$ab = 100$$

Video Solution:



Q16 Text Solution:

$$\begin{aligned} & \frac{1}{\log_2 x} + \frac{1}{\log_4 x} + \frac{1}{\log_8 x} + \dots + \frac{1}{\log_{8192} x} \\ &= \log_x 2 + \log_x 4 + \log_x 8 + \dots \\ &+ \log_x 8192 \\ &= \log_x 2 + \log_x 2^2 + \log_x 2^3 + \dots \\ &+ \log_x 2^{13} \\ &= \log_x 2 + 2 \log_x 2 + 3 \log_x 2 + \dots \\ &+ 13 \log_x 2 \\ &= (1 + 2 + 3 + \dots + 13) \log_x 2 \\ &= \frac{13 \times 14}{2} \log_x 2 \\ &= 91 \log_x 2 \end{aligned}$$

Video Solution:



Q17 Text Solution:

$$\begin{aligned} & \log_{2x-3} (7x^3 - 38x^2 + 67x - 37) = 3 \\ & \Rightarrow 7x^3 - 38x^2 + 67x - 37 = (2x - 3)^3 \\ & \Rightarrow 7x^3 - 38x^2 + 67x - 37 = 8x^3 - 36x^2 \\ &+ 54x - 27 \\ & \Rightarrow x^3 + 2x^2 - 13x + 10 = 0 \end{aligned}$$

To obtain the first root let's try hit and trial with values 1, -1 and so on.

Using $x=1$

$$x^3 + 2x^2 - 13x + 10 = 0$$

So 1 is one of the roots, so now dividing the expression by $(x-1)$

$$\frac{x^3 + 2x^2 - 13x + 10}{x-1} = x^2 + 3x - 10$$

So



$$\begin{aligned}
 x^3 + 2x^2 - 13x + 10 &= 0 \\
 \Rightarrow (x-1)(x^2 + 3x - 10) &= 0 \\
 \Rightarrow (x-1)(x^2 + 5x - 2x - 10) &= 0 \\
 \Rightarrow (x-1)(x+5)(x-2) &= 0 \\
 \Rightarrow x &= 1 \text{ or } 2 \text{ or } -5
 \end{aligned}$$

So

$$2x-3 = -1 \text{ or } 1 \text{ or } -13$$

Since base of a log cannot be negative or 1, so there are no possible values of x.

Video Solution:



Q18 Text Solution:

$$\text{Let } 6^x = 12^y = 72^z = m$$

So

$$x \log 6 = \log m$$

$$\Rightarrow x = \log_6 m$$

Similarly

$$y = \log_{12} m$$

$$\text{and } z = \log_{72} m$$

So

$$\frac{x}{z} - \frac{x}{y}$$

$$= \frac{\log_6 m}{\log_{72} m} - \frac{\log_6 m}{\log_{12} m}$$

$$= (\log_m 72)(\log_6 m)$$

$$- (\log_m 12)(\log_6 m)$$

$$= \log_6 72 - \log_6 12$$

$$\left(\text{Since } \log_a b \times \log_c a = \log_c b \right)$$

$$= \log_6 \left(\frac{72}{12} \right)$$

$$= \log_6 6$$

$$= 1$$

Video Solution:



Q19 Text Solution:

$$\log_{0.5} (3y-4) > \log_{0.25} (64)$$

$$\Rightarrow \log_{0.5} (3y-4) > \log_{(0.5)^2} (8)^2$$

$$\Rightarrow \log_{0.5} (3y-4) > \frac{2}{2} \log_{0.5} 8$$

$$\Rightarrow 3y-4 < 8 \quad \left(\text{If } a \text{ is smaller} \right.$$

$$\left. \text{than 1 and } \log_a b < \log_a c, \text{ then } b > c \right)$$

$$\Rightarrow 3y < 12$$

$$\Rightarrow y < 4$$

also the argument of a log function will be greater than 0

$$\text{So, } 3y-4 > 0$$

$$y > 4/3$$

Hence y can take 2 value, y = 2 and 3 (since y > 0 and y ≠ 1).

So two possible values of y.

Video Solution:



Q20 Text Solution:



$$\begin{aligned}
& \sum_{n=1}^{25} \log_{243} 9^n \\
&= \sum_{n=1}^{25} \log_{3^5} 3^{2n} \\
&= \sum_{n=1}^{25} \left[\frac{2n}{5} (\log_3 3) \right] \\
&= \sum_{n=1}^{25} \frac{2n}{5} \\
&= \frac{2}{5} \left(\sum_{n=1}^{25} n \right) \\
&= \frac{2}{5} (1 + 2 + 3 + \dots + 25) \\
&= \frac{2}{5} \times \frac{25 \times 26}{2} \\
&= 130 \\
&\text{So } x = 130 \\
&5x = 650
\end{aligned}$$

Video Solution:



Q21 Text Solution:

$$\begin{aligned}
& \log_{(x-2)} (x^3 - 8x^2 + 22x - 20) = 2 \\
&\Rightarrow x^3 - 8x^2 + 22x - 20 = (x-2)^2 \\
&\Rightarrow x^3 - 8x^2 + 22x - 20 = x^2 - 4x + 4 \\
&\Rightarrow x^3 - 9x^2 + 26x - 24 = 0 \\
&\text{Checking for first root by hit and trial, and taking values of } x \text{ as } 0, 1, -1, 2 \text{ and so on we will obtain first root as } 2, \text{ so} \\
&(x-2)(x^2 - 7x + 12) = 0 \\
&\Rightarrow (x-2)(x^2 - 3x - 4x + 12) = 0 \\
&\Rightarrow (x-2)(x-3)(x-4) = 0 \\
&\text{So we obtain 3 values of } x \text{ as } 2, 3 \text{ and } 4 \\
&\text{But since the base of the log is } x-2, \text{ so} \\
&x-2 > 0 \\
&\Rightarrow x > 2 \\
&\text{But since the base cannot be } 1, x \text{ cannot be } 3 \\
&\text{So only one value of } x \text{ is possible, i.e., } 4.
\end{aligned}$$

Video Solution:



Q22 Text Solution:

The given 1st equation can be solved as:

$$\begin{aligned}
& \log_5 m = \left(\frac{1}{\log_3 5} + \frac{1}{\log_7 5} + \frac{1}{\log_{19} 5} \right) \\
& \log_5 m = \log_5 3 + \log_5 7 + \log_5 19 \\
& \log_5 m = \log_5 (3 \times 7 \times 19) = \log_5 (399) \\
& m = 399 \\
& \text{Now,}
\end{aligned}$$

$$\log_{10} \log_2 (5^{\sqrt{5x+1}} + m) = 1$$

$$\log_2 (5^{\sqrt{5x+1}} + m) = 10$$

$$5^{\sqrt{5x+1}} + m = 2^{10} = 1024$$

$$5^{\sqrt{5x+1}} + 399 = 1024 \quad [\text{Since, } m = 399]$$

$$5^{\sqrt{5x+1}} = 625 = 5^4$$

$$\sqrt{5x+1} = 4$$

$$5x+1 = 16$$

$$5x = 15$$

$$x = 3$$

Video Solution:



Q23 Text Solution:

The given equation is

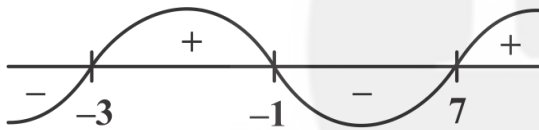


$$\begin{aligned}
& \log_7(x-1) - 2\log_7(x+2) \\
& + \log_7(x-5) < \log_7[31 - x(x-18)] \\
& \log_7(x-1) + \log_7(x+2) + \log_7(x-5) \\
& < \log_7[31 - x(x-18)] \\
& \log_7 \left[\frac{(x-1)(x+2)(x-5)}{31-x(x-18)} \right] < 0 \\
& \frac{(x-1)(x+2)(x-5)}{31-x(x-18)} < 1 \\
& (x-1)(x+2)(x-5) - 31 + x(x-18) \\
& < 0
\end{aligned}$$

Simplifying the inequality, we get

$$\begin{aligned}
& x^3 - 4x^2 - 7x + 10 + x^2 - 18x - 31 < 0 \\
& x^3 - 3x^2 - 25x - 21 < 0 \\
& x^2(x+1) - 4x(x+1) - 21(x+1) < 0 \\
& (x+1)(x^2 - 4x - 21) < 0 \\
& (x+1)(x^2 - 7x + 3x - 21) < 0 \\
& (x+1)(x+3)(x-7) < 0
\end{aligned}$$

Using the Wavy-Curve Method, we get



$$x \in (-\infty, -3) \cup (-1, 7)$$

But, the given equation involves logarithmic functions and the input for logarithmic functions must be greater than 0.

So, the given equation to exist, we must take $x > 5$.

Therefore, only one integer value of x can be taken, i.e., $x = 6$

Video Solution:



Q24 Text Solution:

The given equation is:

$$\begin{aligned}
& \log_9(x-2) = \log_3(x-4) \\
& \log_{3^2}(x-2) = \log_3(x-4) \\
& \frac{1}{2}\log_3(x-2) = \log_3(x-4) \\
& (x-2)^{\frac{1}{2}} = (x-4) \\
& x-2 = x^2 - 8x + 16 \\
& x^2 - 9x + 18 = 0 \\
& (x-3)(x-6) = 0 \\
& x = 3, 6
\end{aligned}$$

But, at $x = 3$, the function $\log_3(x-4)$ does not exist.

So, $x = 6$ is the only solution.

Video Solution:



Q25 Text Solution:

$$\begin{aligned}
& (\log_b a \cdot \log_c a - 1) + (\log_a b \cdot \log_c b - 1) \\
& + (\log_a c \cdot \log_b c - 1) = 0 \\
& \Rightarrow \frac{\log a}{\log b} \times \frac{\log a}{\log c} - 1 + \frac{\log b}{\log a} \times \frac{\log b}{\log c} - 1 \\
& + \frac{\log c}{\log a} \times \frac{\log c}{\log b} - 1 = 0 \\
& \Rightarrow \frac{(\log a)^2}{\log b \cdot \log c} + \frac{(\log b)^2}{\log a \cdot \log c} + \frac{(\log c)^2}{\log b \cdot \log a} = 3 \\
& \Rightarrow \frac{(\log a)^3 + (\log b)^3 + (\log c)^3}{\log a \cdot \log b \cdot \log c} = 3 \\
& \Rightarrow (\log a)^3 + (\log b)^3 + (\log c)^3 \\
& = 3(\log a \cdot \log b \cdot \log c) \\
& \Rightarrow \log a + \log b + \log c = 0 \\
& (a^3 + b^3 + c^3 = 3abc \Rightarrow a + b + c = 0) \\
& \Rightarrow \log(abc) = 0 \\
& \Rightarrow abc = 1
\end{aligned}$$

Video Solution:




Q26 Text Solution:

$$\text{Numerator} = \log_4 y - 3$$

$$\text{For } y < 64, \text{ numerator} < 0$$

$$\text{For } y = 64, \text{ numerator} = 0$$

$$\text{For } y > 64, \text{ numerator} > 0$$

$$\text{Denominator} = 4 - \log_3 y$$

$$\text{For } y < 81, \text{ denominator} > 0$$

$$\text{For } y = 81, \text{ denominator} = 0$$

$$\text{For } y > 81, \text{ denominator} < 0$$

So the given expression will be non negative for $64 \leq y < 81$

So there are 17 such integral values of y

Video Solution:

Q27 Text Solution:

$$\log_x \frac{x^2+9}{x^2-7} = 2$$

$$\Rightarrow \frac{x^2+9}{x^2-7} = x^2$$

$$\Rightarrow x^2 + 9 = x^4 - 7x^2$$

$$\Rightarrow x^4 - 8x^2 - 9 = 0$$

$$\Rightarrow (x^2 - 9)(x^2 + 1) = 0$$

$$\Rightarrow x^2 = 9 \text{ or } -1$$

$$\Rightarrow x^2 = \pm 3 \text{ or } \pm \sqrt{-1}$$

Since $\pm \sqrt{-1}$ are not real, they will not be considered

Also base of log cannot be negative, so 3 is the only possible value of x . So only one real value of x satisfies the equation.

Video Solution:

Q28 Text Solution:

For the logarithmic function to be defined, its argument must be greater than 0, i.e.,

$$\left[\frac{9x - 2 \left(3^{\log_3 x + \log_3 x^{\frac{1}{2}} + \log_3 x^{\frac{1}{4}} + \dots} \right)}{7} \right] > 0$$

$$9x - 2 \left(3^{\log_3 x + \log_3 x^{\frac{1}{2}} + \log_3 x^{\frac{1}{4}} + \dots} \right) > 0$$

$$9x - 2 \cdot 3^{\log_3 x} \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right) > 0$$

$$9x - 2 \cdot 3^{\log_3 x} \left(\frac{1}{1 - \frac{1}{2}} \right) > 0$$

$$9x - 2 \cdot 3^{\log_3 x^2} > 0$$

$$9x - 2x^2 > 0$$

$$x \in \left(0, 4.5 \right) \dots \dots \dots \left(i \right)$$

We have,

$$x^2 + 2x - 8 \leq 0$$

$$x^2 - 2x + 4x - 8 \leq 0$$

$$x(x - 2) + 4(x - 2) \leq 0$$

$$(x + 4)(x - 2) \leq 0$$

$$\text{So, } -4 \leq x \leq 2 \dots (ii)$$

Using (i) and (ii), we have

$$0 < x \leq 2.$$

So, the integer values of 'x' that satisfy will be 1 and 2, thus the sum of such values will be 3.

Video Solution:


**Q29 Text Solution:**

$$5^x = 3025$$

$$\Rightarrow x \left(\log 5 \right) = 2 \left(\log 55 \right)$$

$$\Rightarrow x = \frac{2(\log 55)}{\log 5}$$

Similarly

$$y = \frac{2(\log 55)}{\log 11}$$

So

$$xy = \frac{2(\log 55)}{\log 5} \times \frac{2(\log 55)}{\log 11} = \frac{4(\log 55)^2}{\log 5 \times \log 11}$$

Also

$$\begin{aligned} x + y &= \frac{2(\log 55)}{\log 5} + \frac{2(\log 55)}{\log 11} \\ &= \frac{2(\log 55)(\log 11 + \log 5)}{\log 5 \times \log 11} = \frac{2(\log 55)^2}{\log 5 \times \log 11} \end{aligned}$$

So

$$\frac{3xy}{x+y} = \frac{\frac{3 \times 4(\log 55)^2}{\log 5 \times \log 11}}{\frac{2(\log 55)^2}{\log 5 \times \log 11}} = 6$$

Video Solution:**Q30 Text Solution:**

The given expression is:

$$\log_{(x+7)} [(x+5)/(x+3)]$$

First of all, $x + 7 > 0$

$$\text{i.e., } x > -7 \dots (i)$$

Second, $x + 7 \neq 1$

$$\text{i.e., } x \neq -6 \dots (ii)$$

Third, $(x+5)/(x+3) \neq 0$

$$\text{i.e., } x \neq -5 \dots (iii)$$

Fourth, $x + 3 \neq 0$

$$\text{i.e., } x \neq -3 \dots (iv)$$

Fifth, $(x+5)/(x+3) > 0$ Multiplying both sides by $(x+3)^2$, we get

$$(x+3)(x+5) > 0$$

Using the Wavy-Curvy Method, we get



$$x < -5 \text{ and } x > -3 \dots (v)$$

Combining all 5 conditions, we get the following negative integer values of x :

$$x = -2, -1$$

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